A Methodology to Optimize Hardware Resources for Distributed Neural Network Inference on FPGAs

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Introduction

FPGAs are not suited for large scale DNNs or Convolution Neural Networks, due to the massive number of floating point operations they require. To address this problem, multiple techniques have been developed in the past years within frameworks like FINN.

Our work focuses on split neural network inference across multiple low-power FPGA devices. We propose an automated method for neural network partitioning combined with FINN folding optimization.

FINN is an experimental open-source framework that converts an ONNX (Open Neural Network Exchange) model of a Deep Neural Network (DNN) into an FPGA implementation.

Methodology

Network Splitting

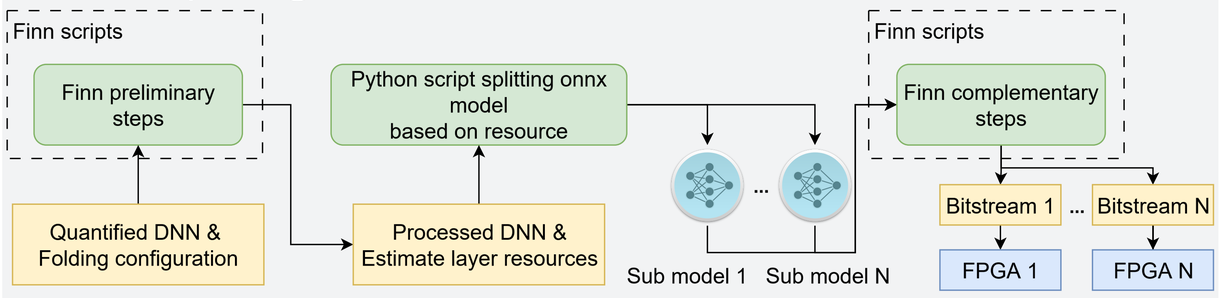


Figure 1: Full process from base model ONNX to ready to use bitstreams.

System Design

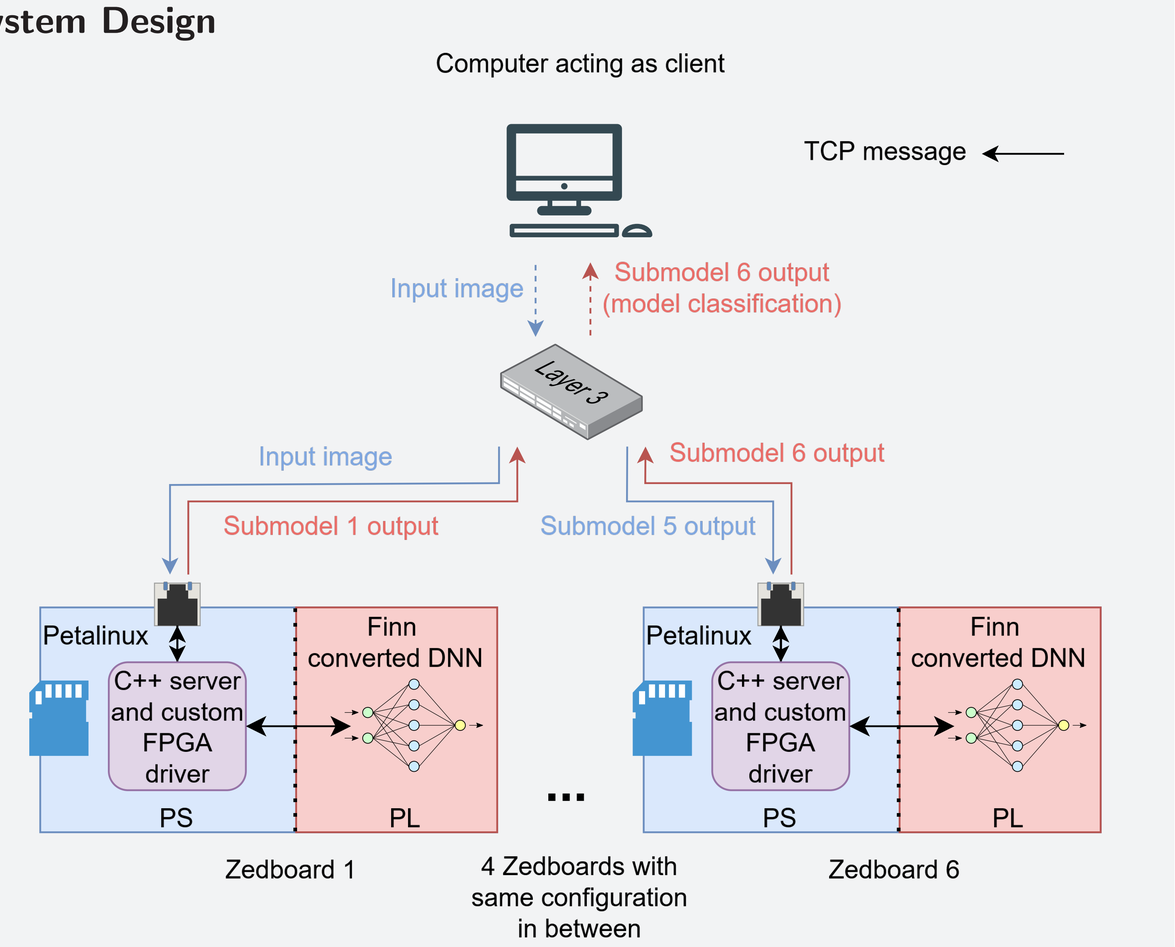


Figure 2: Example of a full communication for a DNN divided into six bitstreams, from an image input to the model classification (Sub model 6 output), from right to left in a sequential manner.

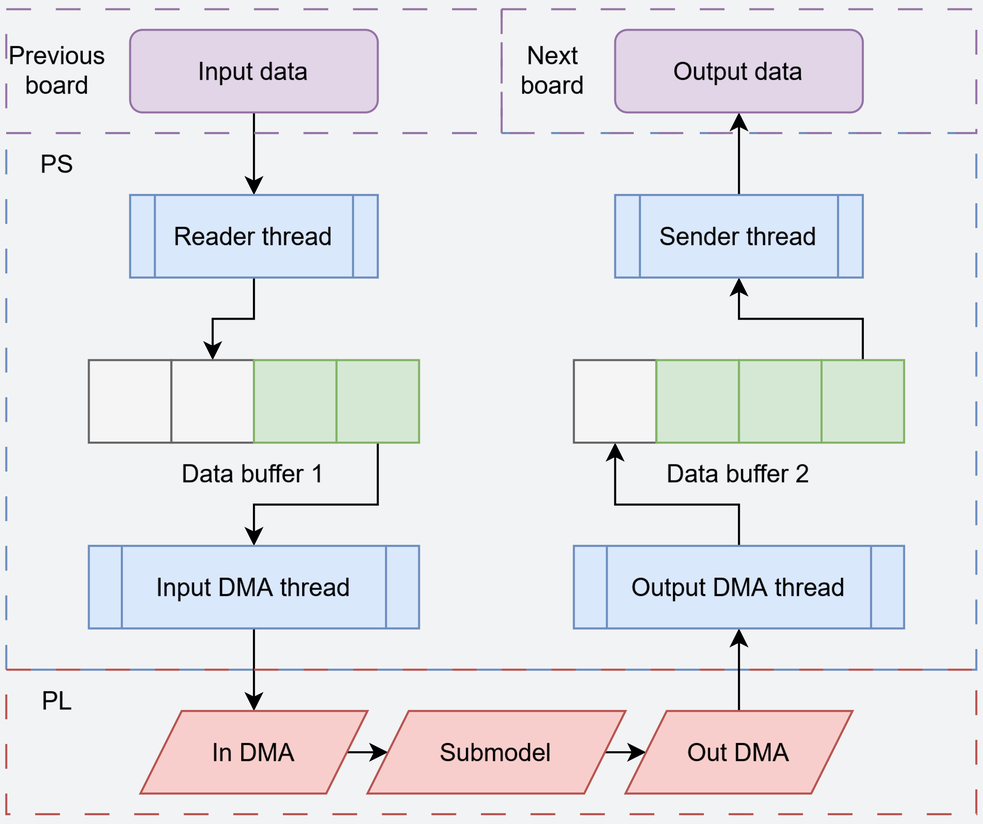


Figure 3: Illustration of the software architecture and data flow on a board from the input of the previous board (or client) to the next. The data buffers are FIFO, their occupancy here is purely illustrational.

Optimization

Byte packing

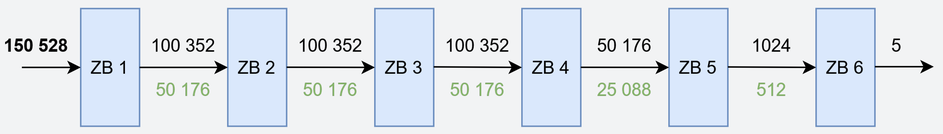


Figure 4: Transfer size in byte between each board for the MobileNetV1. In black, transfer size before byte packing; in green transfer size after byte packing.

Cut Nudging

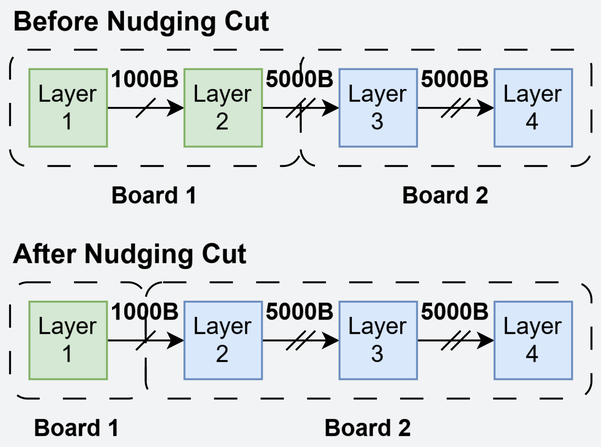


Figure 5: Illustration of the "nudging" of a cut, moving layer 2 to board 2 to reduce network transfer from 5000 bytes to 1000 bytes leaving larger transfer on the PL.

Experimental Results

Estimated results

Table 1: Expected maximum throughput estimated by FINN estimator for the CNVW2A2 and MobileNetV1, in frame per second

Hardware Comp.	Model	Est. FPS	Total Est. Power (W)	Search Time
2 Zedboards	CNVW2A2	664.3	4.274	0:00:18
6 Zedboards	MBNet V1	622.8	18.990	0:15:14
4 Zed. + 1 ZC706	MBNet V1	622.8	20.167	1:06:08
6 Zed. + 1 ZC706	MBNet V1	1122.7	24.745	0:55:28

Experimental results

Table 2: Comparison to existing works on Tiny ML acceleration

Work	Hardware	Nb boards	Peak FPS	Est. power (W)
Previous	**Zedboards**	6	118.3	**16.176**
Ours	**Zedboards**	6	272.4	**16.629**
Ours	**ZC706 + Zedboards**	**7**	260.6	22.035
[1]	Alveo U280	2	**5923**	152.3
[2]	Zedboard	1	22.6*	N/A

Conclusion

System Features

- Split inference of DNN across several off-the-shelf SoC-FPGA boards
- Automated split and folding algorithm
- Reasonable compromise between throughput and power consumption

Future Work

- Extend the search space with alternative hardware layer configurations to better exploit underutilized FPGA resources (*e.g.*, LUTRAM, BRAM)

Publications

1. Mathieu Hannoun et al. "Inférence distribuée pour modèle de deep learning sur SoC-FPGA". In: *GRETSI 2025, 30° Colloque sur le traitement du signal et des images*. Aug. 2025
2. Mathieu Hannoun et al. "Towards a Distributed Quantized Machine Learning Inference Using Commodity SoC-FPGAs Using FINN". In: *Works in Progress in Embedded Computing Journal* 11.1 (2025). DOI: 10.64552/wipiec.v11i1.76
3. Mathieu Hannoun et al. "Folding-Nets : A Methodology to Optimize Hardware Resources for Distributed Neural Network Inference on FPGAs". In: *Submitted, available on demand* (2026)

References

[1] Tobias Alonso et al. "Elastic-DF: Scaling Performance of DNN Inference in FPGA Clouds through Automatic Partitioning". In: *ACM Trans. Reconfigurable Technol. Syst.* (2021). DOI: 10.1145/3470567.

[2] Alexander Montgomerie-Corcoran et al. "SAMO: Optimised Mapping of Convolutional Neural Networks to Streaming Architectures". In: 2022. DOI: 10.1109/FPL57034.2022.00069.